Results Overview

• Designs generated: 1000

• Designs passing all filters: 0

• MMCIF files of final designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox2/final_ranked_designs/final_20_designs

• Metrics and sequences of 20 final designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox2/final_ranked_designs/final_designs_metrics_20.csv

• Metrics of all designs are in:

/home/ayushregmibagale/gbm-protein-switch/boltzgen/output/sox2/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
-- boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb`

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 20 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by α = 0.001:

• α = 0 → 100 % quality focus (same as Top set)

• α = 1 → 100 % diversity focus (ignores quality)

• Quality – composite of metrics such as iPTM, salt-bridges, ΔSASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.

• Diversity – 1 − sequence identity between designs.

We use α = 0.001, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
design_sequence	designed amino acids (may be subset of chain)
designed_chain_sequence	full sequence of the chain containing designed residues (recommended for synthesis)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_iptm	predicted TM score for design–target contacts (higher = better)
design_to_target_iptm	same as design_iptm but for multi-chain designs
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	ΔSASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table)then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	1000
filter_rmsd	True	2.5	0
filter_rmsd_design	True	2.5	724
designfolding-filter_rmsd	True	2.5	821
ALA_fraction	True	0.3	805
GLY_fraction	True	0.2	990
GLU_fraction	True	0.2	618
LEU_fraction	True	0.3	999
VAL_fraction	True	0.2	998

Sorting Criteria

Metric	Inverse Importance
design_to_target_iptm	1
design_ptm	1
neg_min_design_to_target_pae	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2

Results Summary

Aggregate statistics

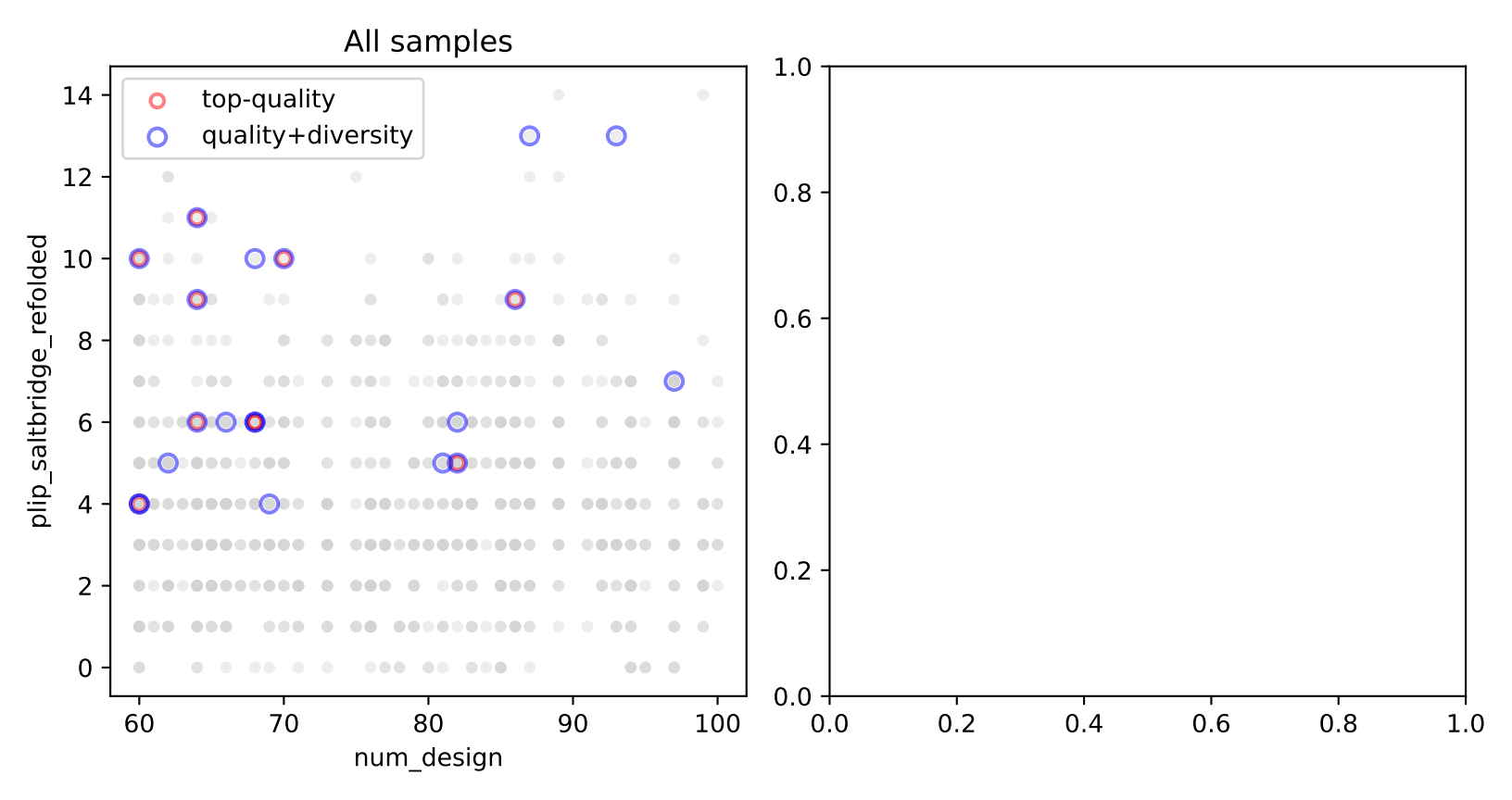
Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 20 quality+diversity designs produced by the lazy-greedy selection.

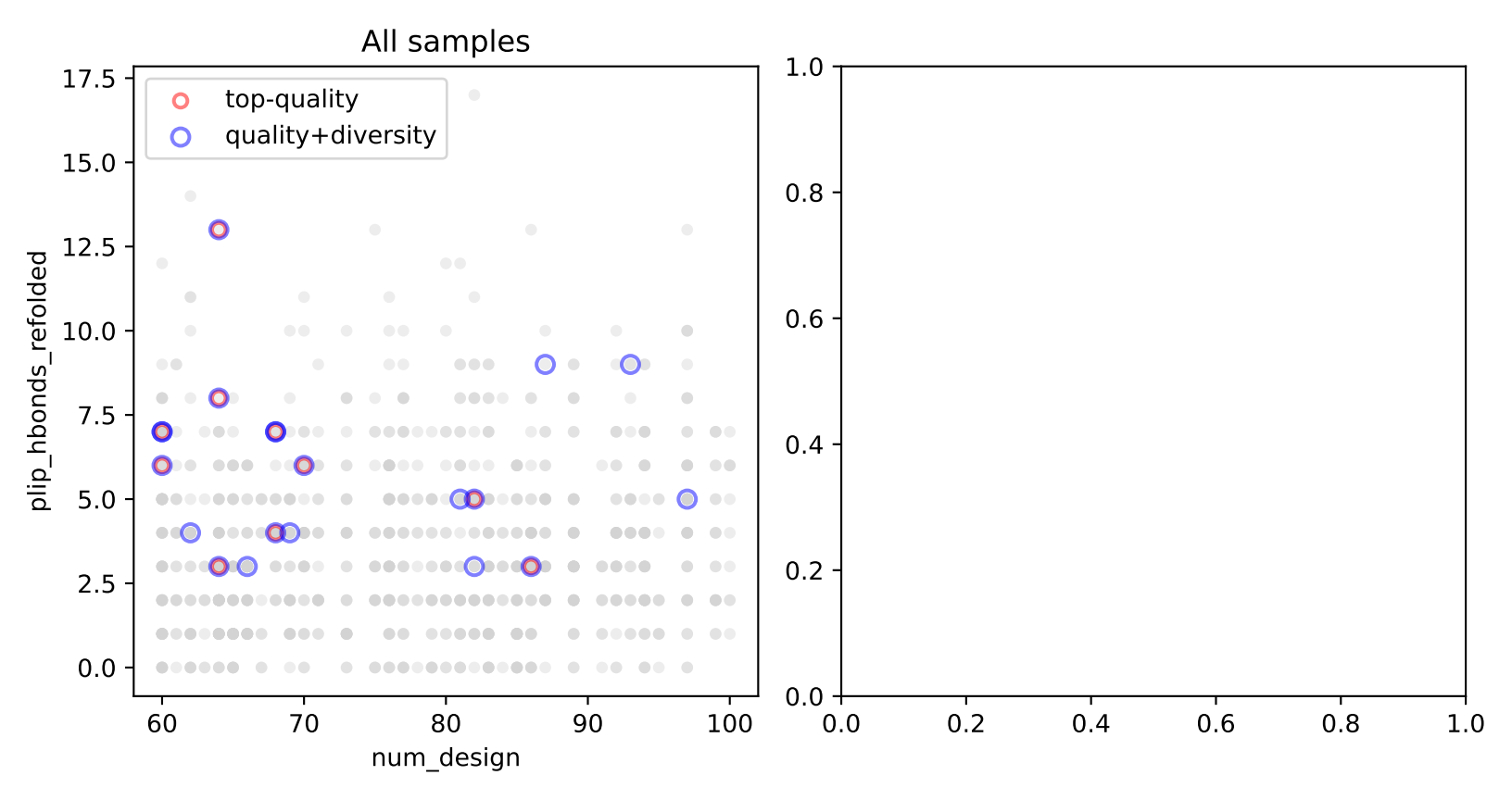
Metric	Mean	Mean top 10	Mean top 20 diverse
Num designs	1000	-	-
num_design	77.530	68.600	72.550
filter_rmsd	16.703	17.141	16.775
designfolding-filter_rmsd	1.856	0.949	0.937
design_iptm	0.151	0.240	0.227
design_to_target_iptm	0.151	0.240	0.227
min_design_to_target_pae	15.859	11.419	11.962
delta_sasa_refolded	971.574	1200.908	1161.159
plip_saltbridge_refolded	4.174	7.600	7.450
plip_hbonds_refolded	3.644	6.200	5.900

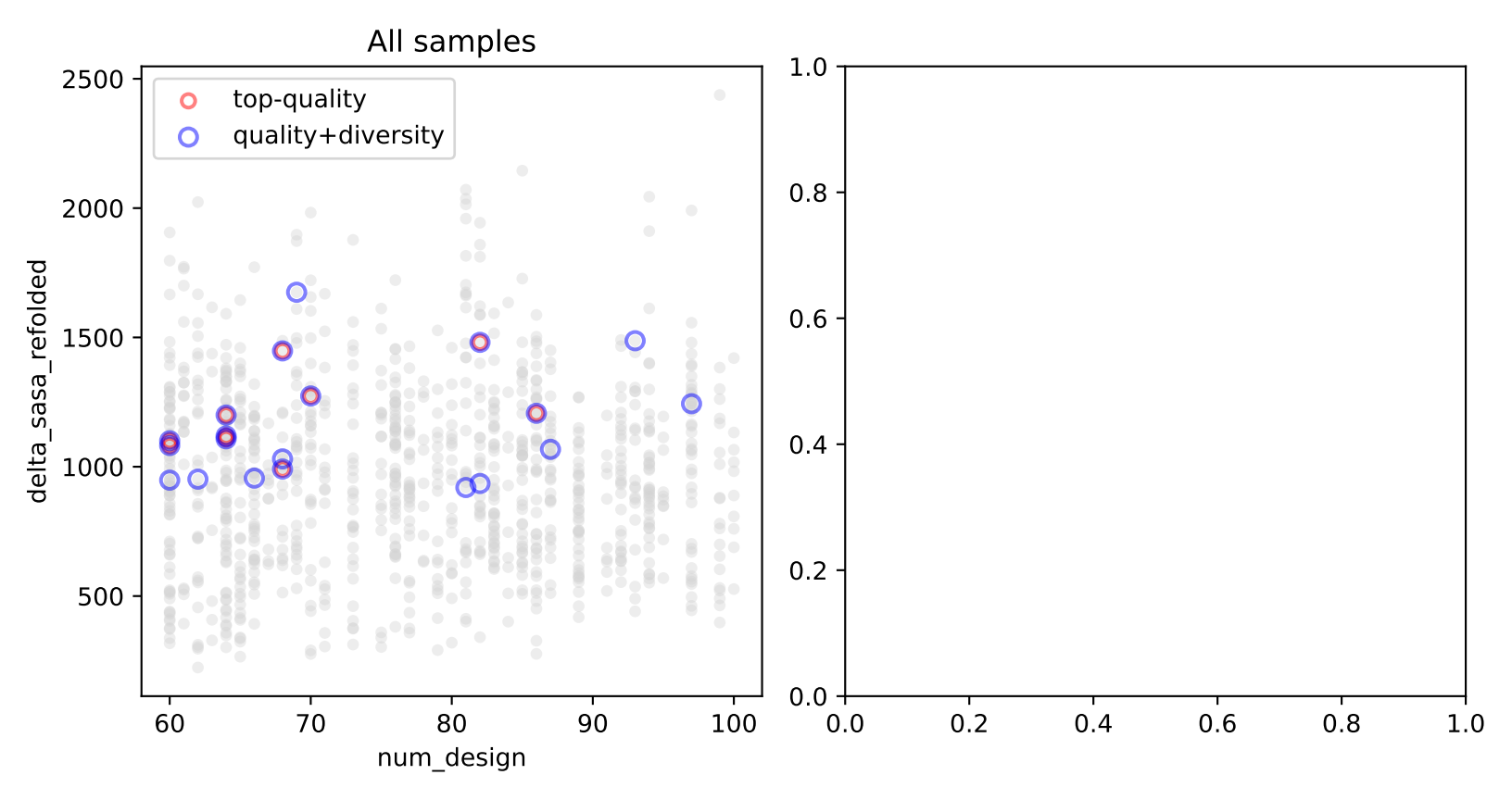
Scatter Plots

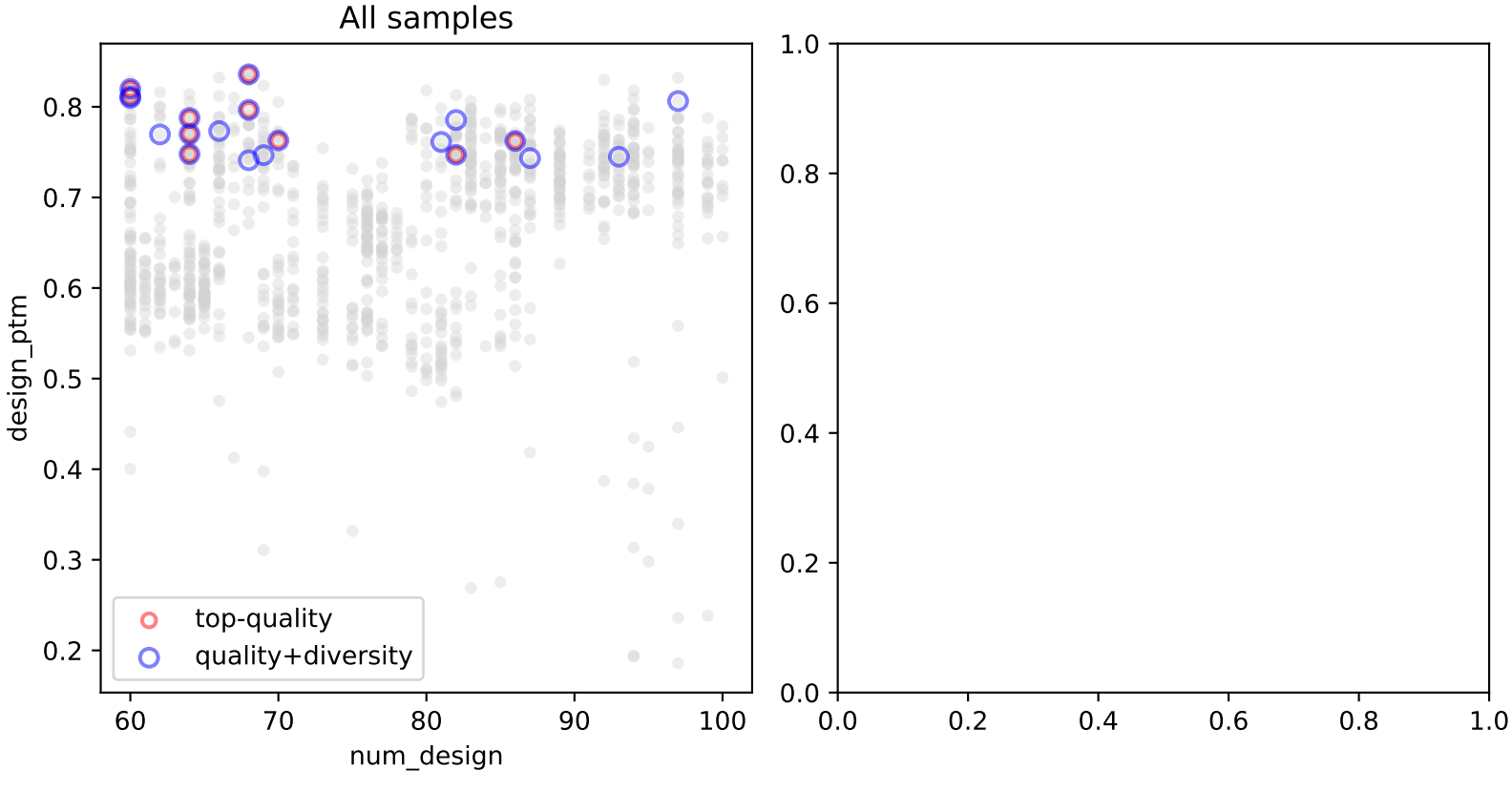
Metric relationships

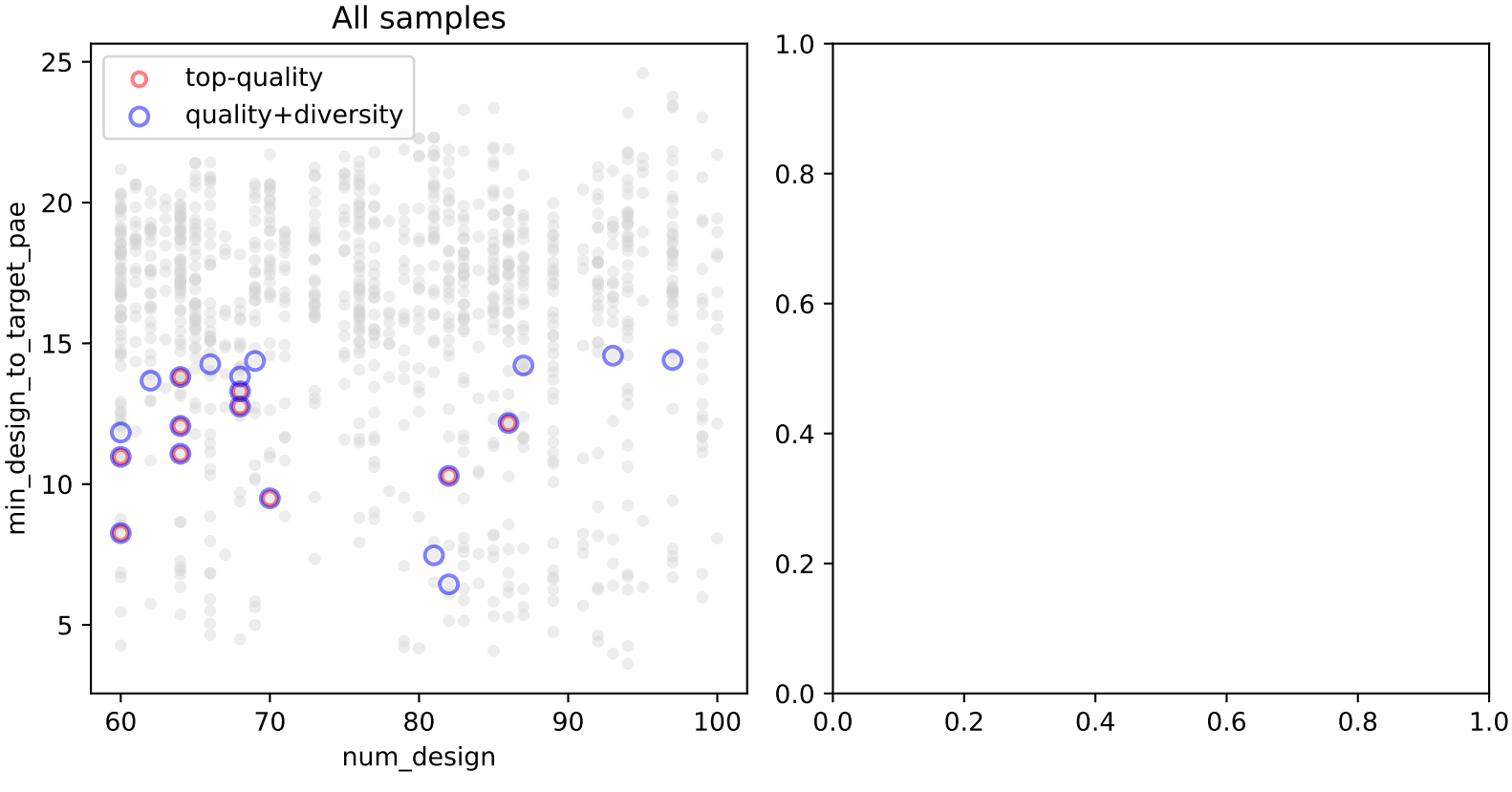
Each scatter page contains two panels:
• Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
• Right – same but limited to designs passing the RMSD filter.

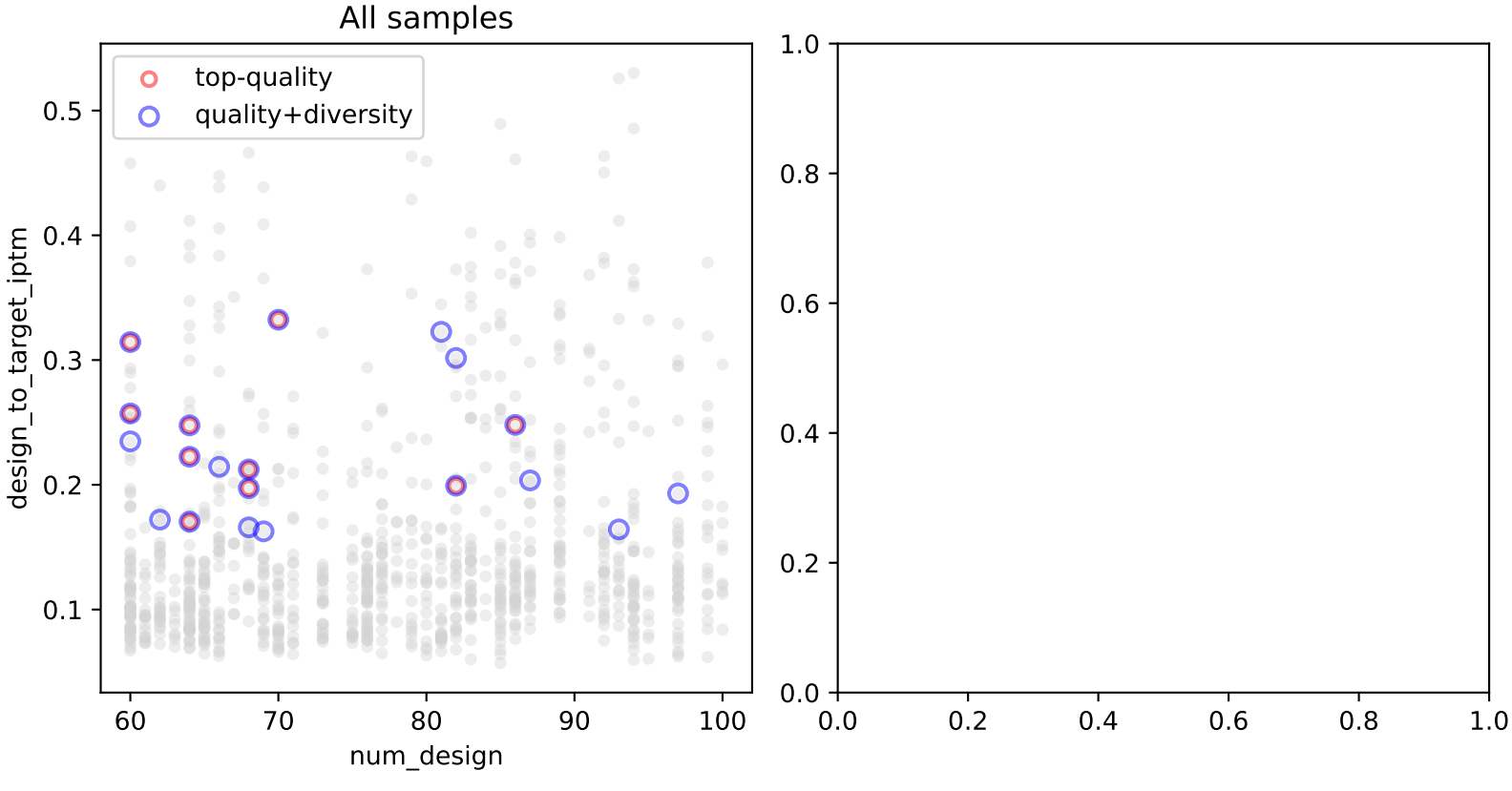

All samples
top-quality
quality+diversity
plip_saltbridge_refolded
num_design
0
2
4
6
8
10
12
14
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0

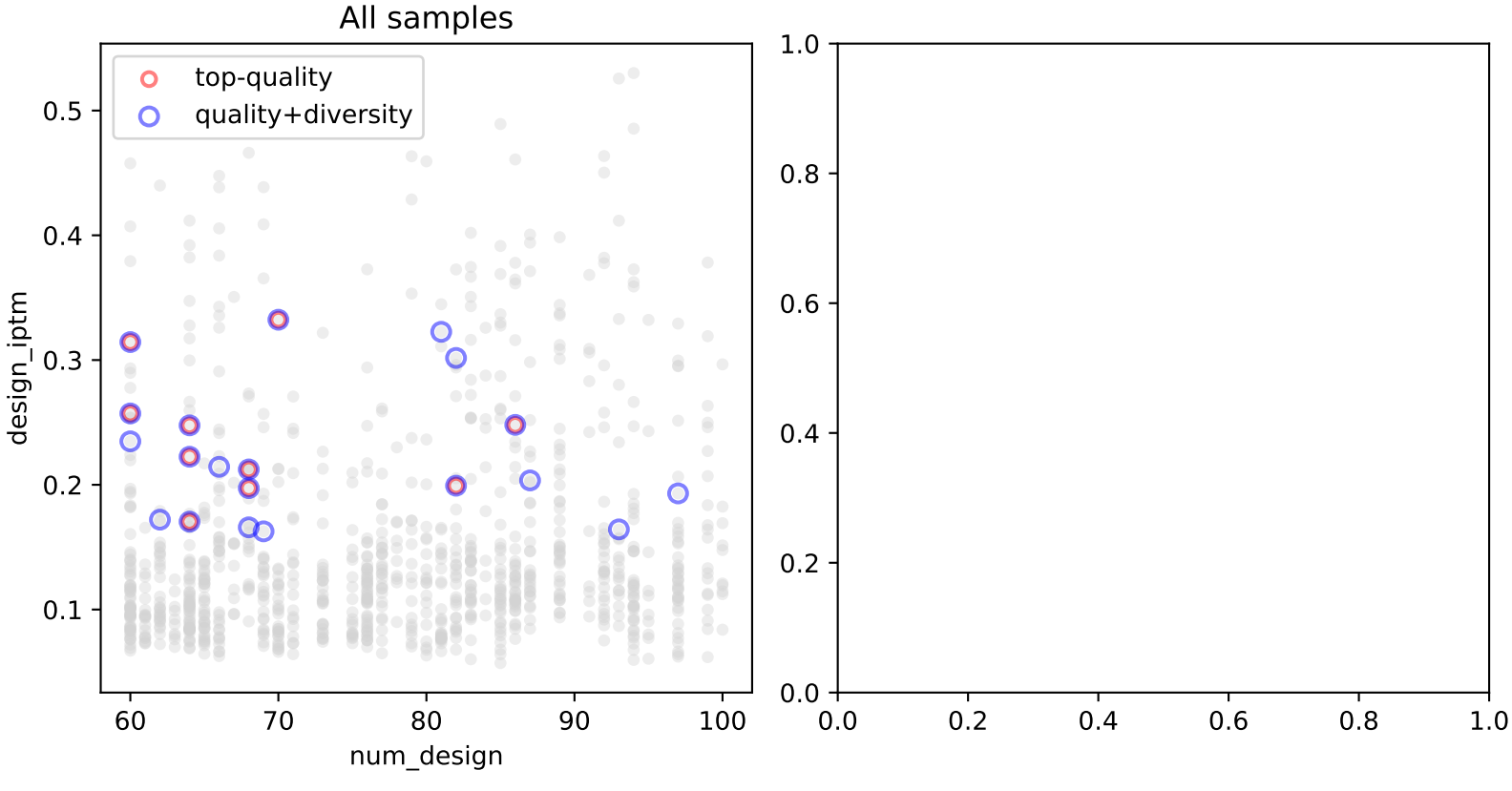

All samples
top-quality
quality+diversity
plip_hbonds_refolded
num_design
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
60
70
80
90
100
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

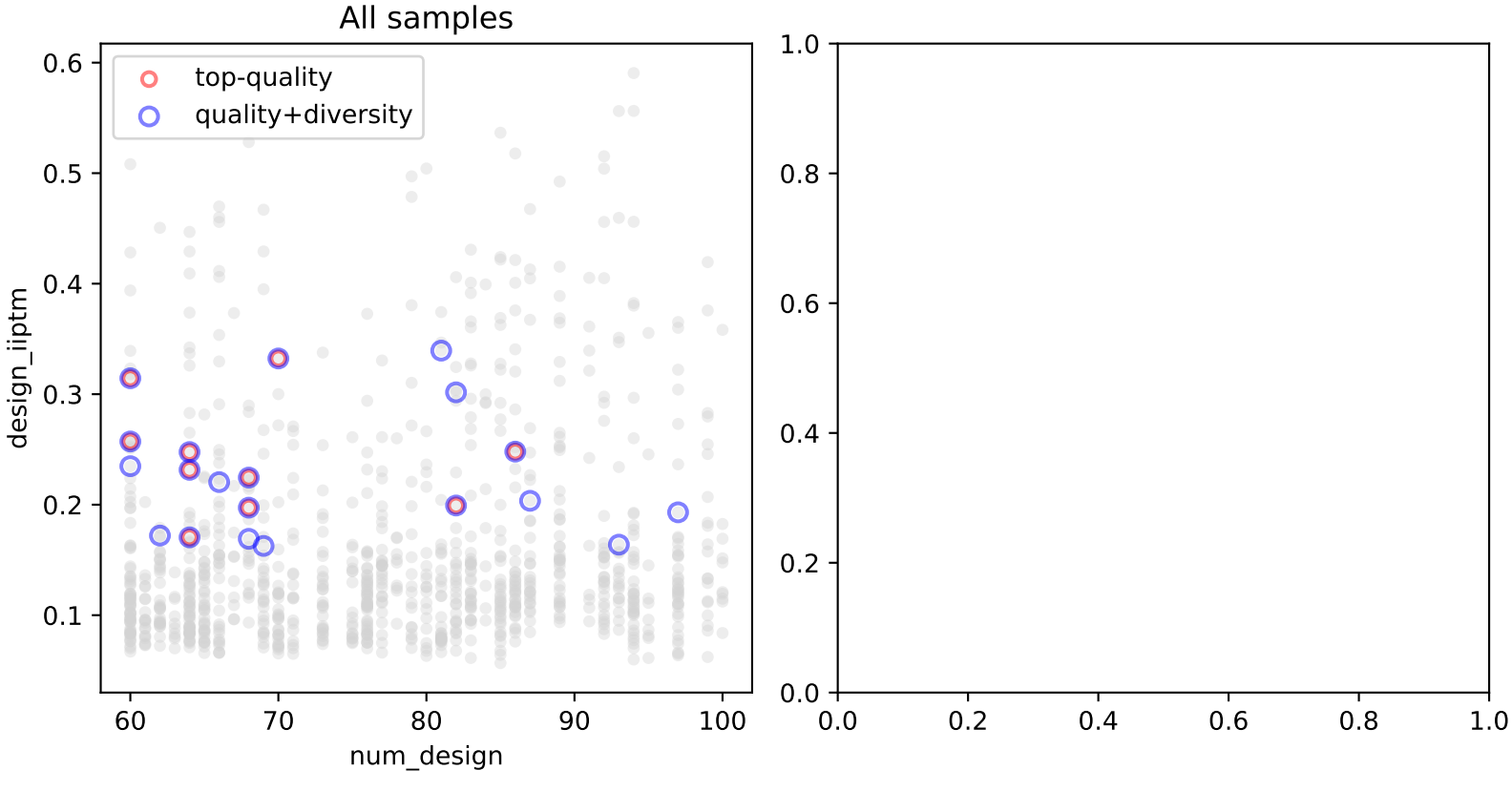

All samples
top-quality
quality+diversity
delta_sasa_refolded
num_design
2500
2000
1500
1000
500
60
70
80
90
100
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

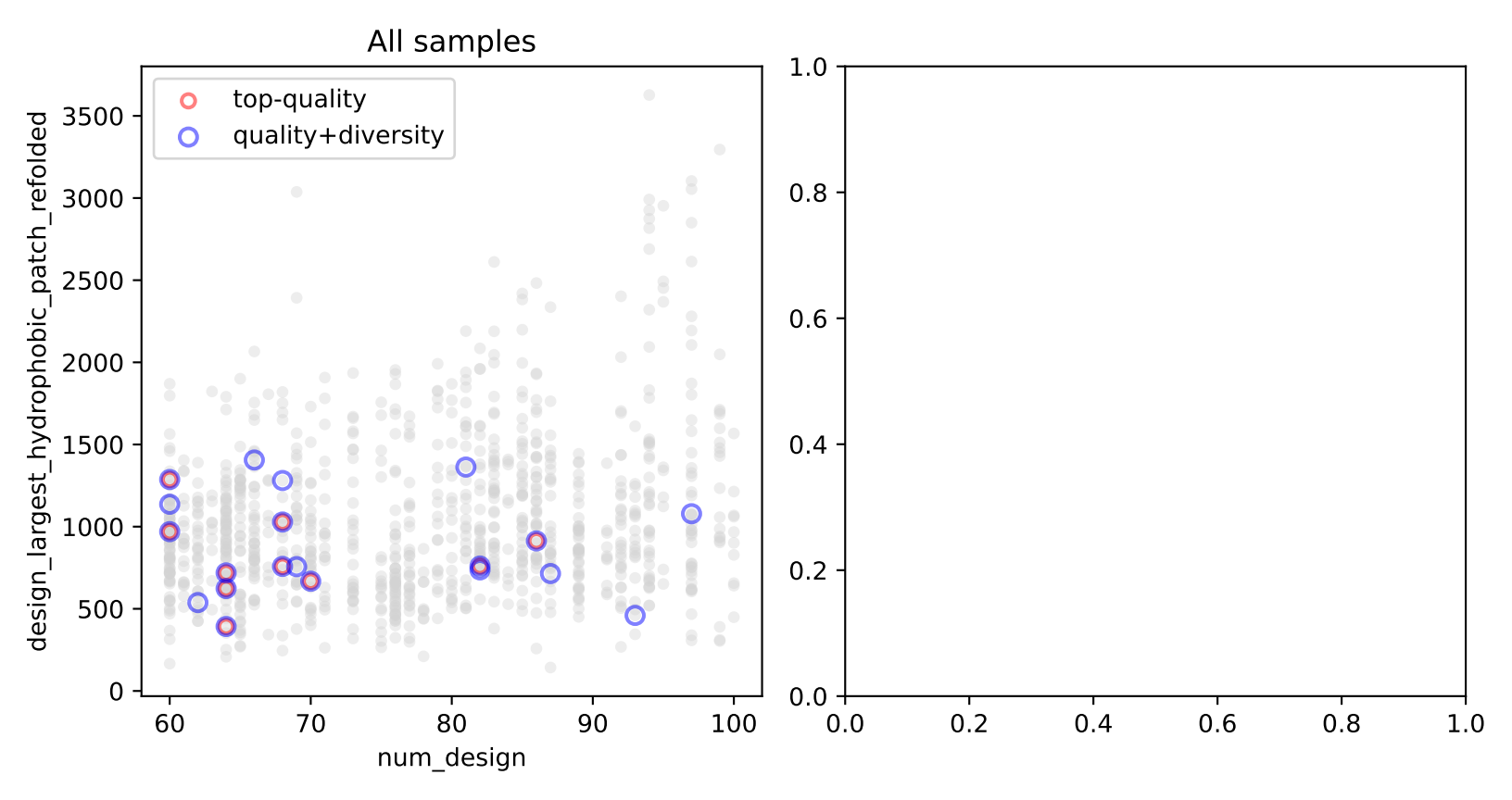

All samples
design_ptm
num_design
top-quality
quality+diversity
0.2
0.3
0.4
0.5
0.6
0.7
0.8
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0

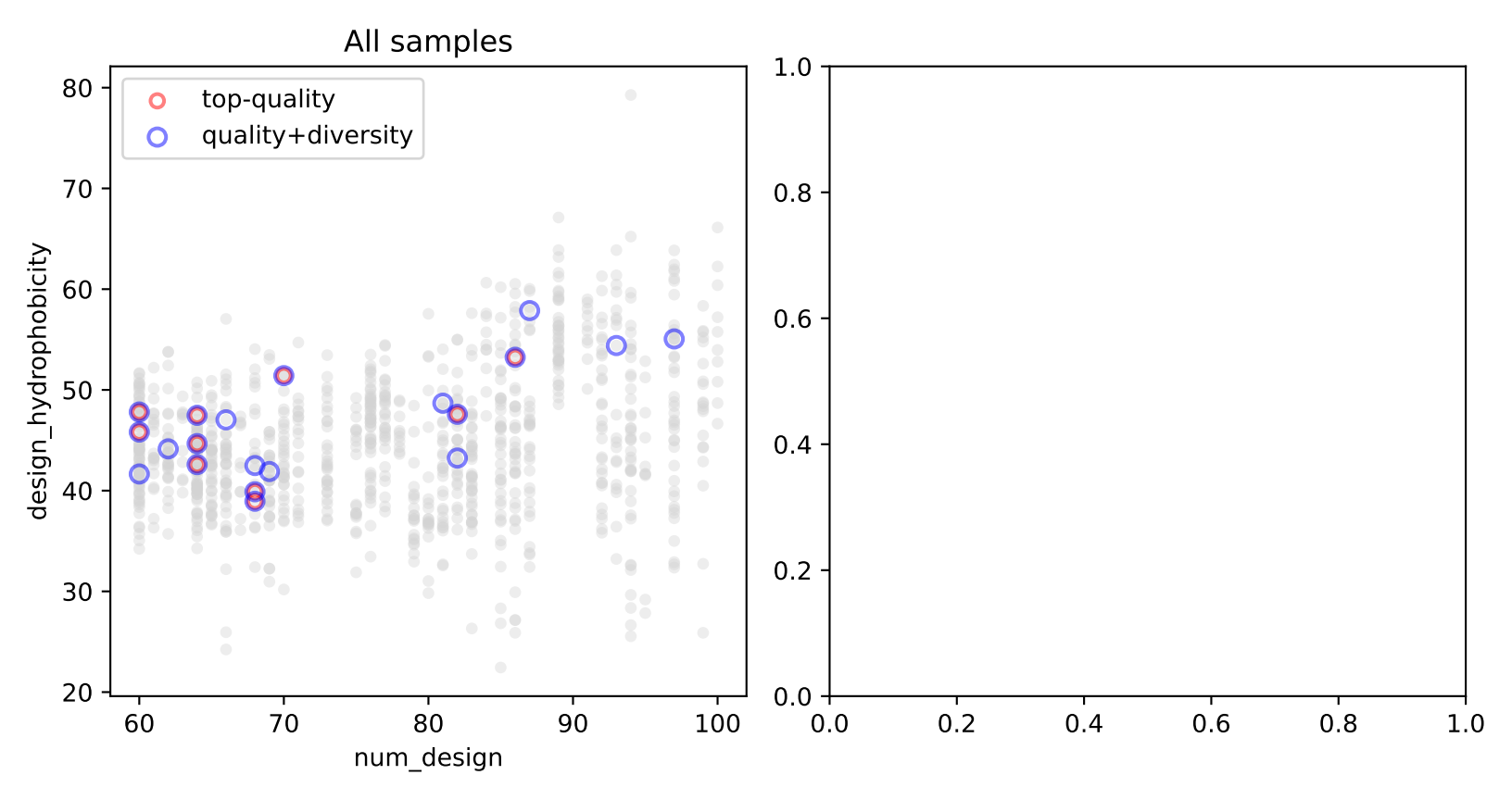

All samples
top-quality
quality+diversity
min_design_to_target_pae
num_design
25
20
15
10
5
60
70
80
90
100
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


All samples
top-quality
quality+diversity
design_to_target_iptm
num_design
0.1
0.2
0.3
0.4
0.5
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0


All samples
top-quality
quality+diversity
design_iptm
num_design
0.1
0.2
0.3
0.4
0.5
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0


All samples
top-quality
quality+diversity
design_iiptm
num_design
0.6
0.5
0.4
0.3
0.2
0.1
60
70
80
90
100
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

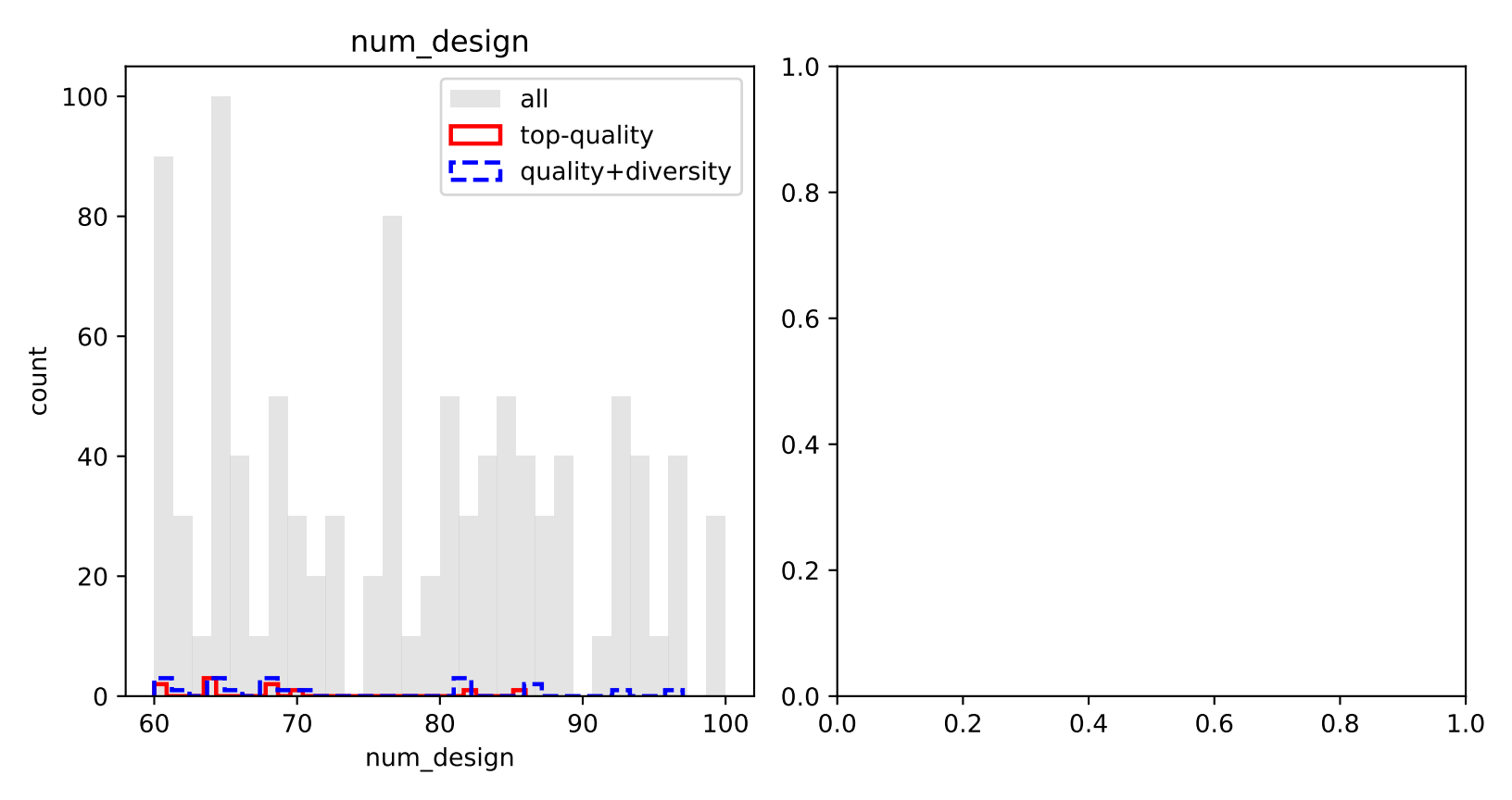

All samples
top-quality
quality+diversity
design_largest_hydrophobic_patch_refolded
num_design
0
500
1000
1500
2000
2500
3000
3500
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0

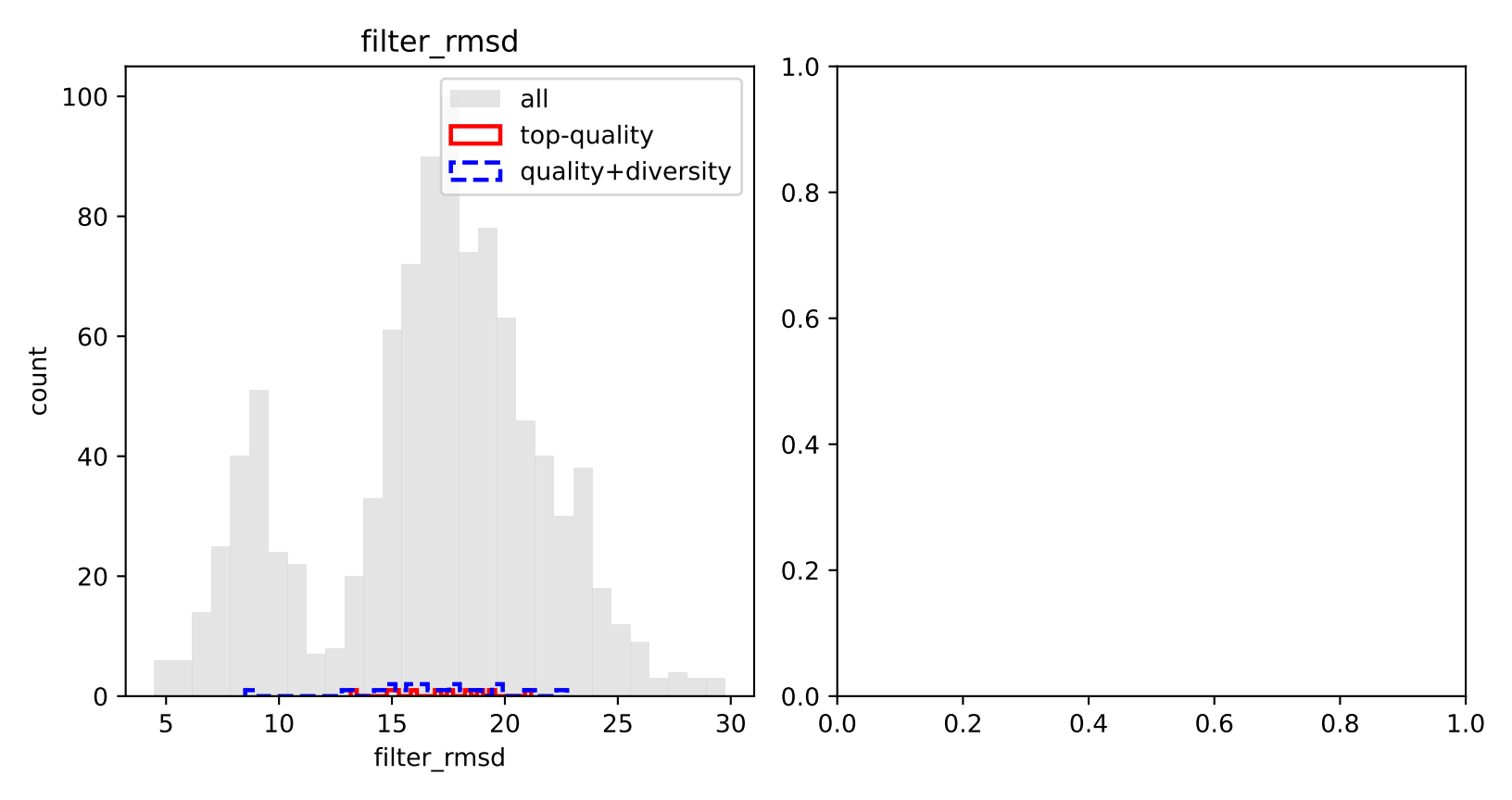

All samples
top-quality
quality+diversity
design_hydrophobicity
num_design
80
70
60
50
40
30
20
60
70
80
90
100
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

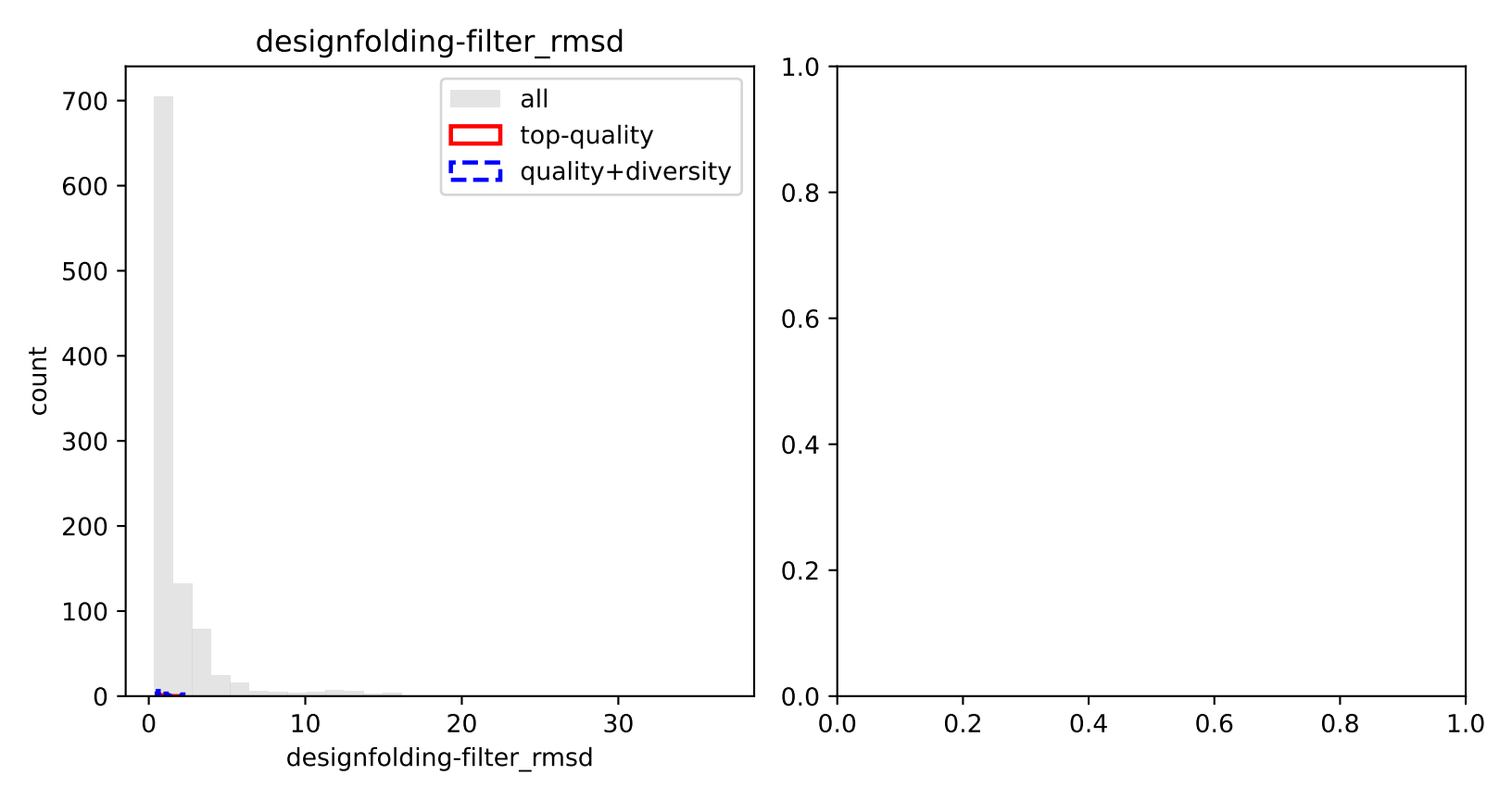
Metric Distributions

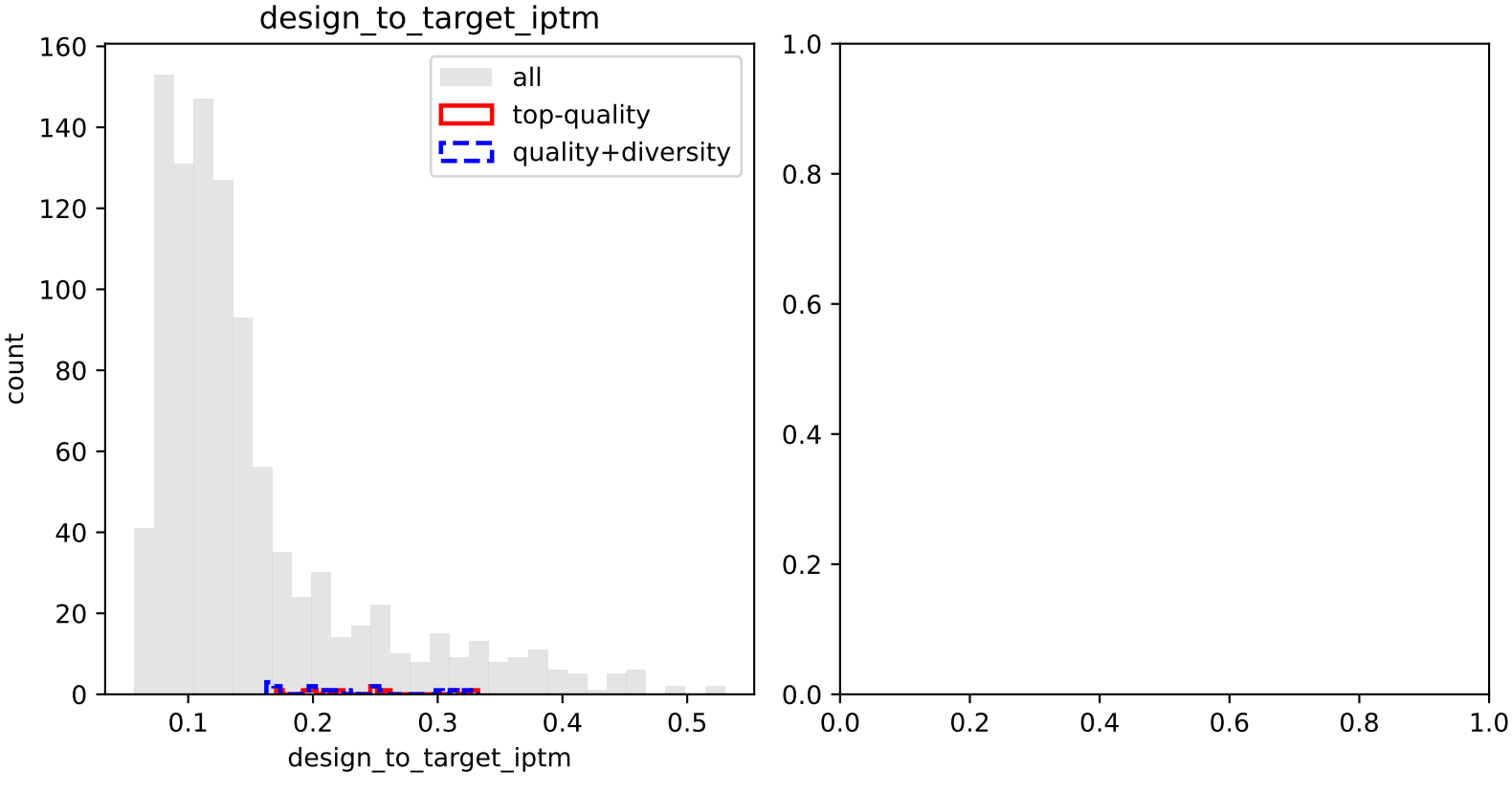
Histograms

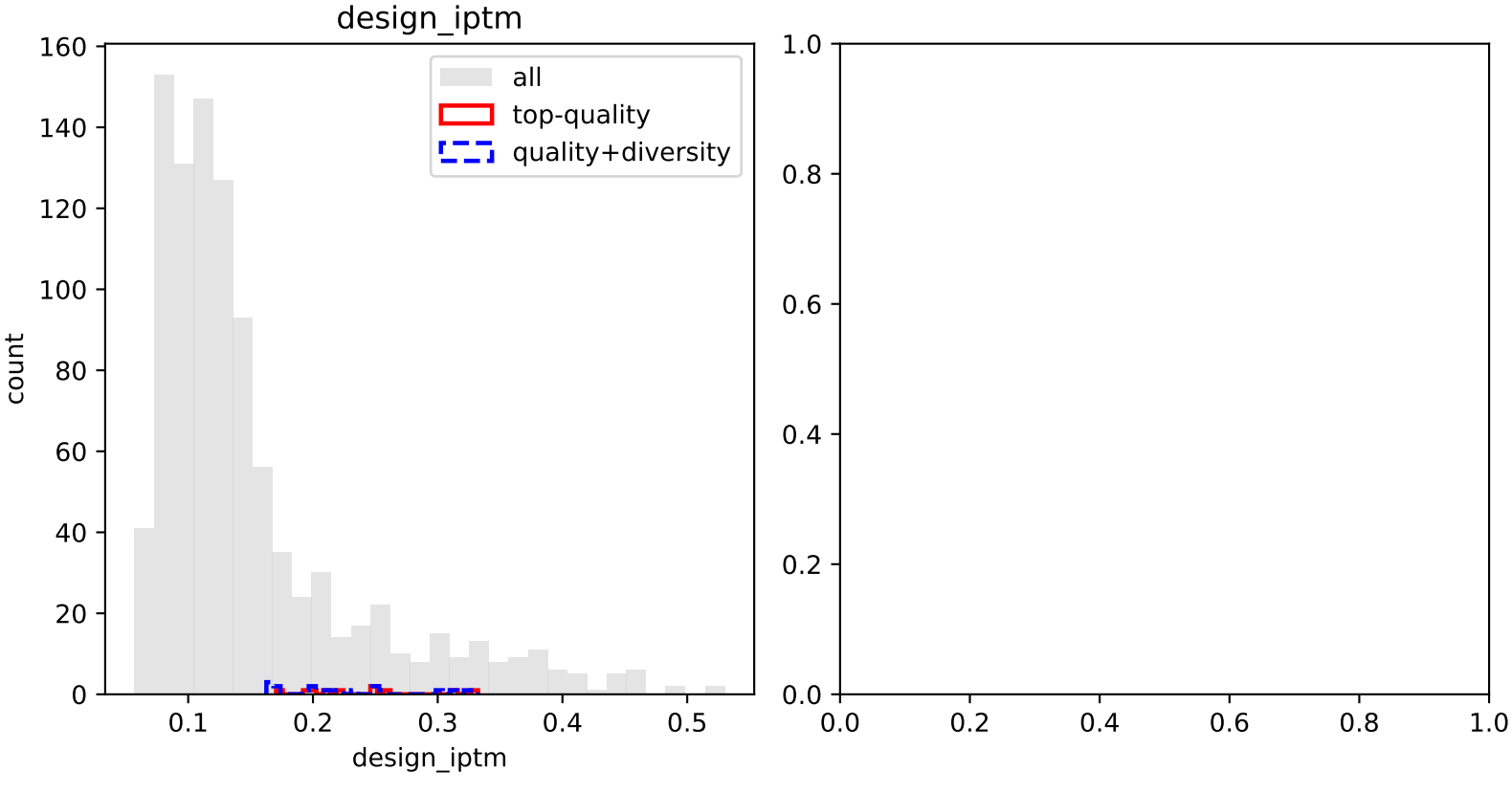
Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

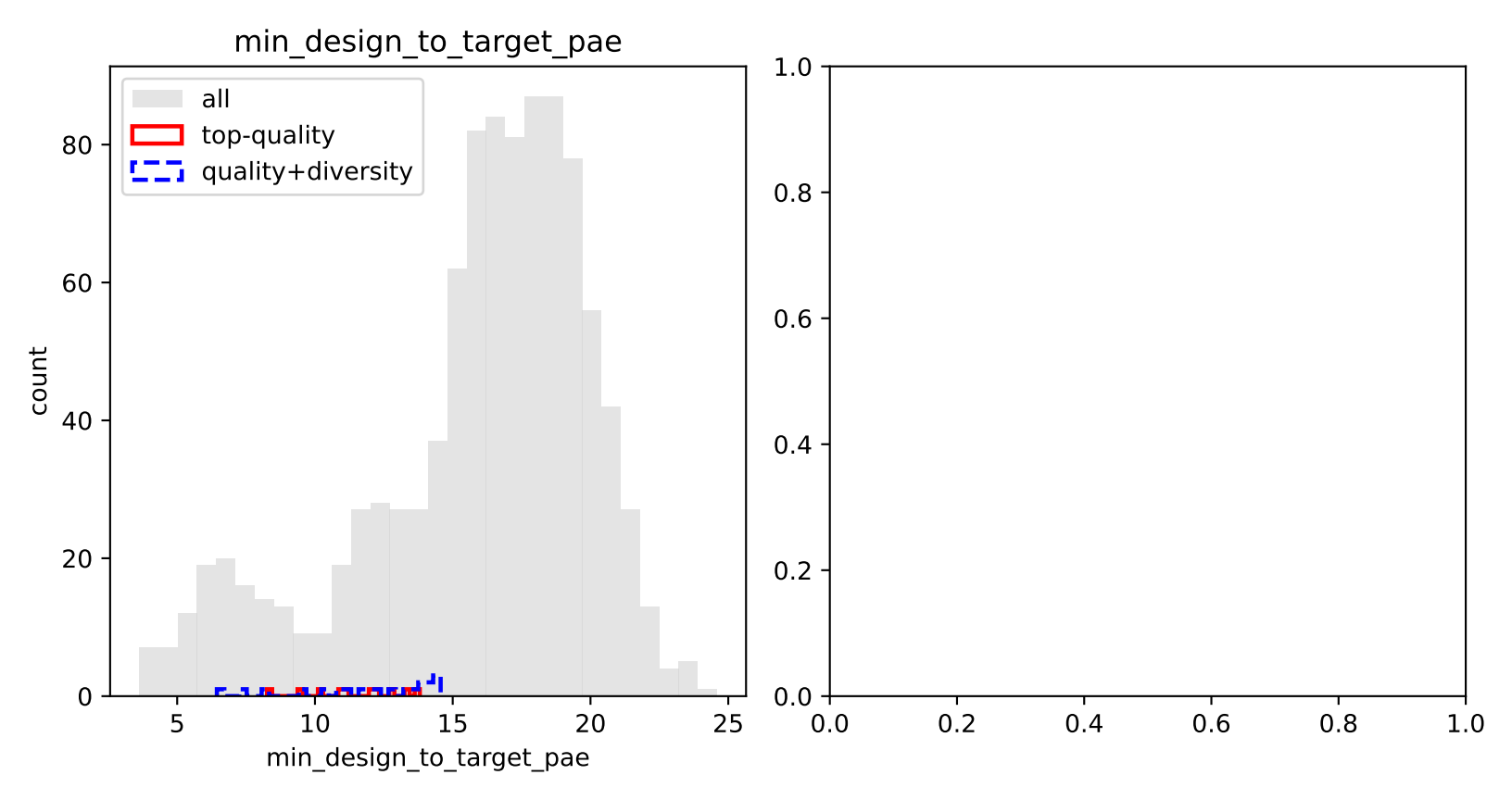

num_design
all
top-quality
quality+diversity
count
num_design
0
20
40
60
80
100
60
70
80
90
100
0.0
0.2
0.4
0.6
0.8
1.0

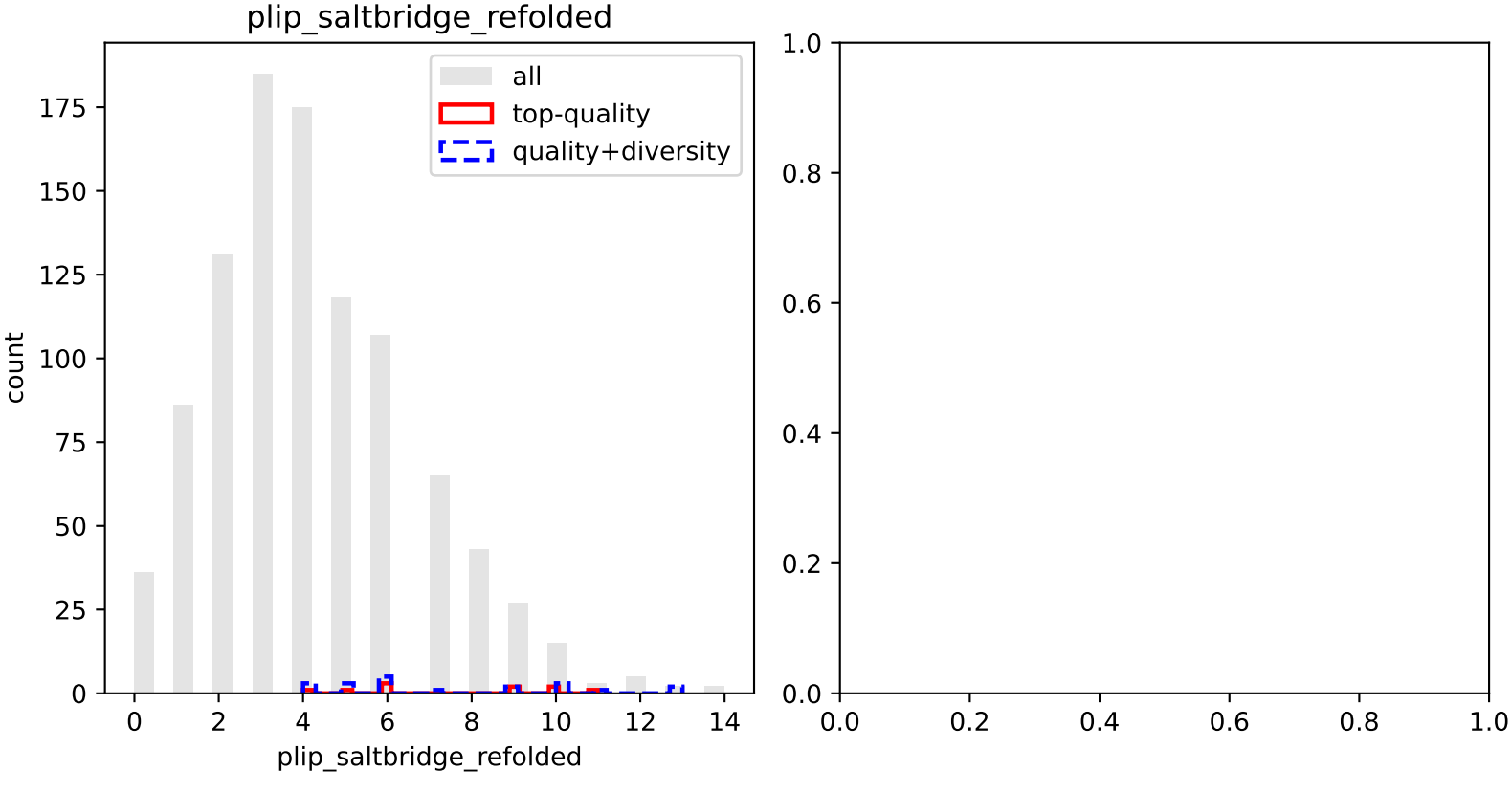

filter_rmsd
all
top-quality
quality+diversity
count
filter_rmsd
0
20
40
60
80
100
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
1.0

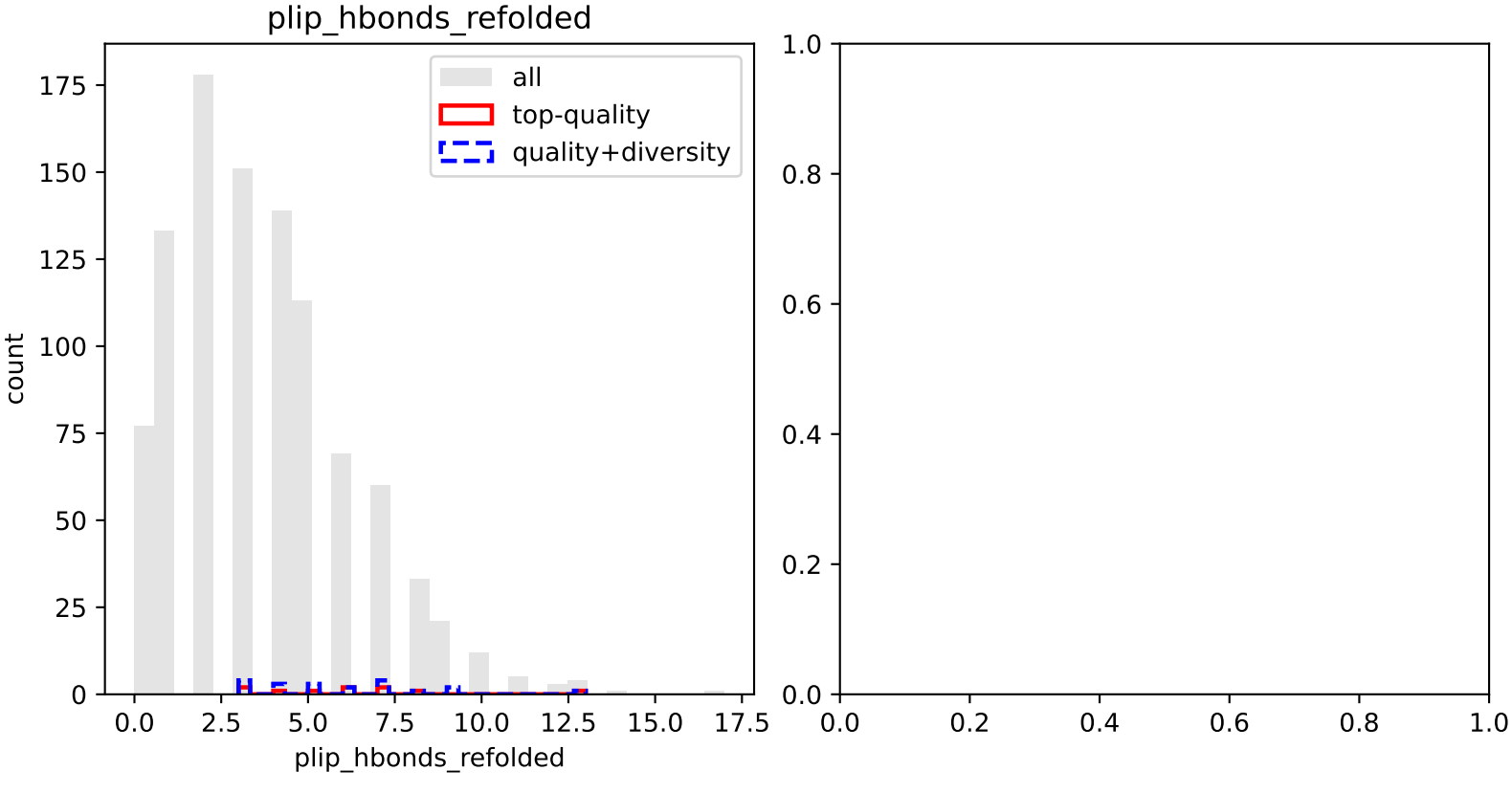

designfolding-filter_rmsd
all
top-quality
quality+diversity
count
designfolding-filter_rmsd
0
100
200
300
400
500
600
700
0
10
20
30
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0

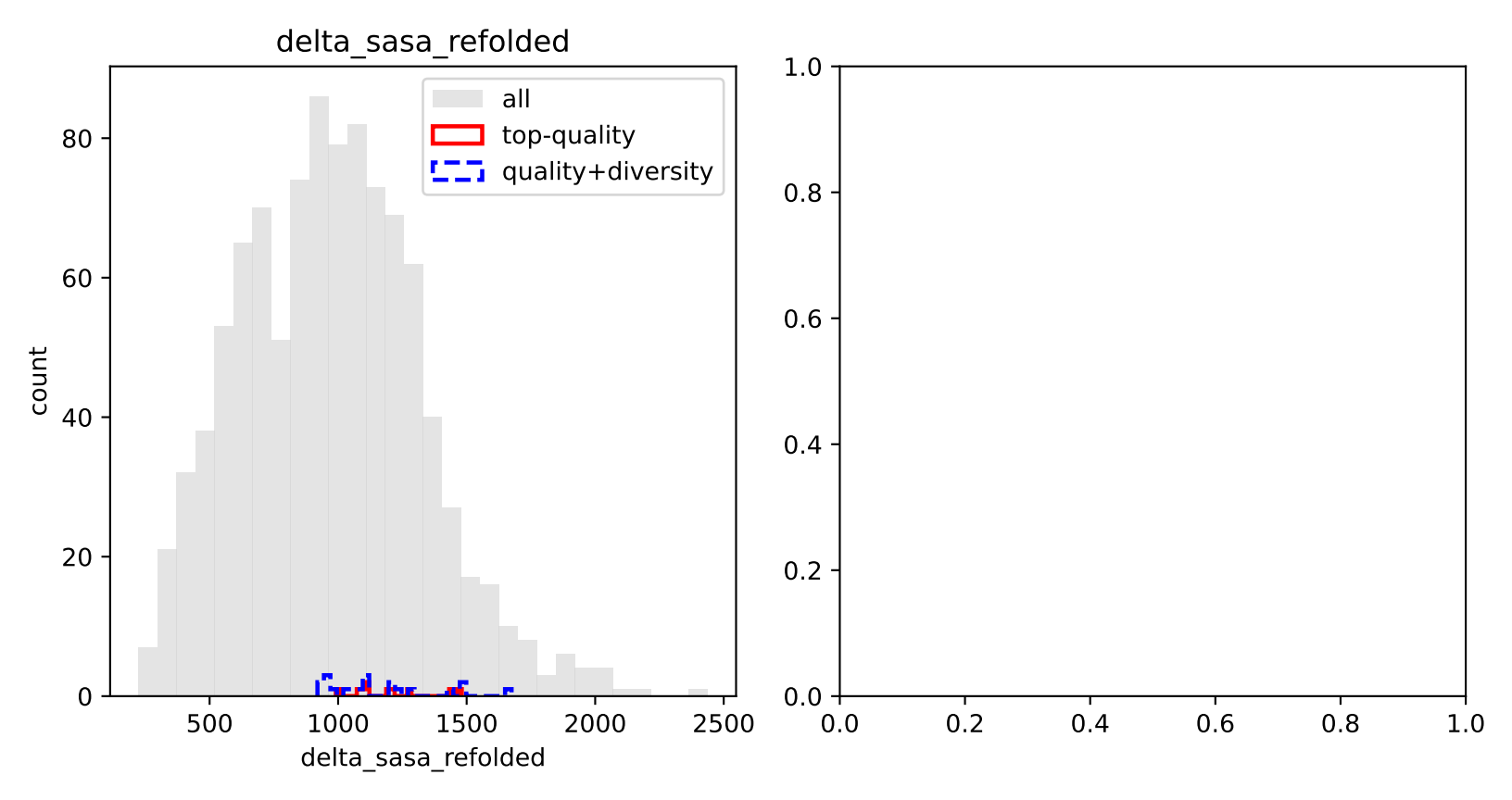

design_to_target_iptm
all
top-quality
quality+diversity
count
design_to_target_iptm
0
20
40
60
80
100
120
140
160
0.1
0.2
0.3
0.4
0.5
0.0
0.2
0.4
0.6
0.8
1.0

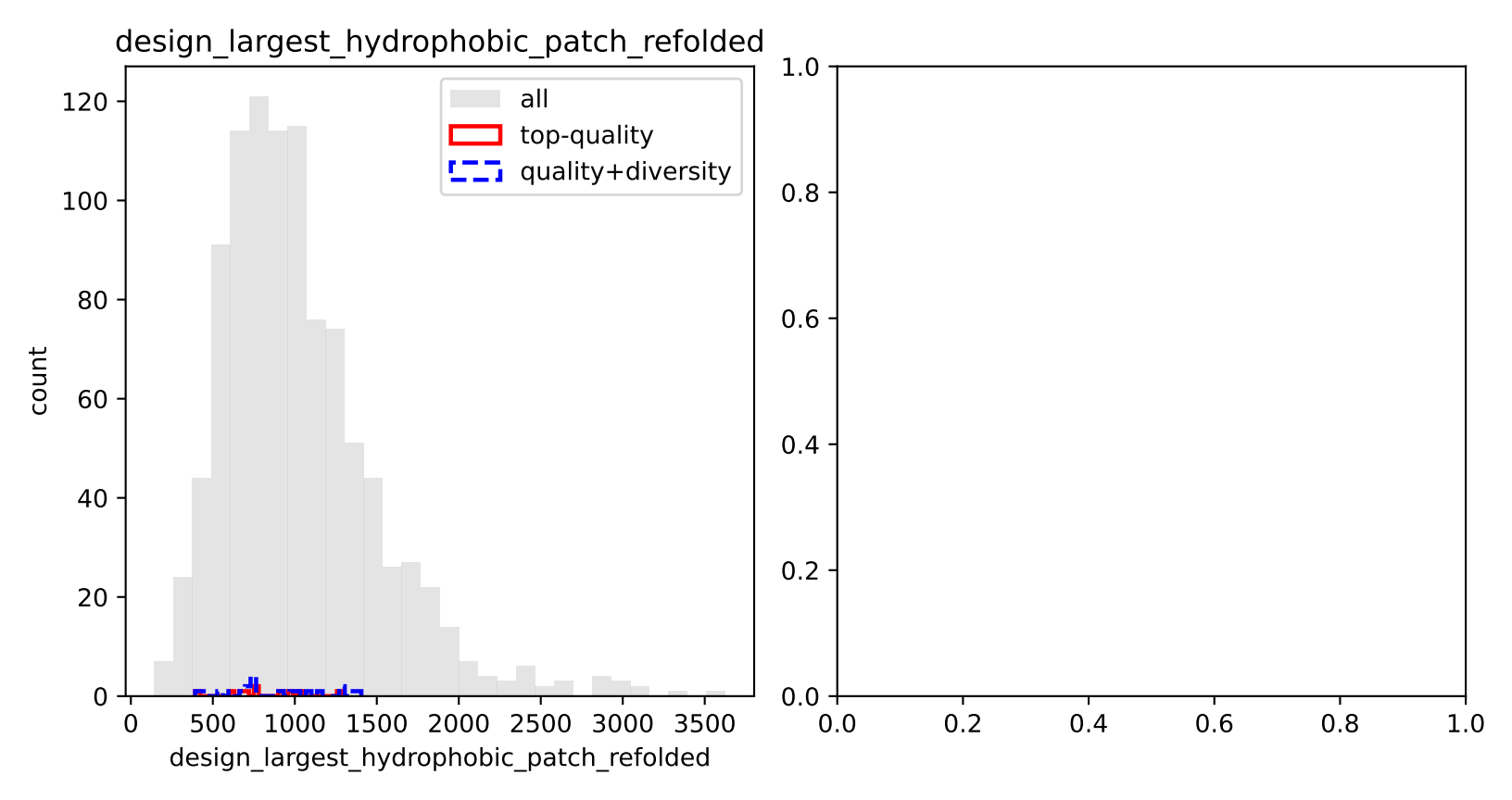

design_iptm
all
top-quality
quality+diversity
count
design_iptm
0
20
40
60
80
100
120
140
160
0.1
0.2
0.3
0.4
0.5
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0

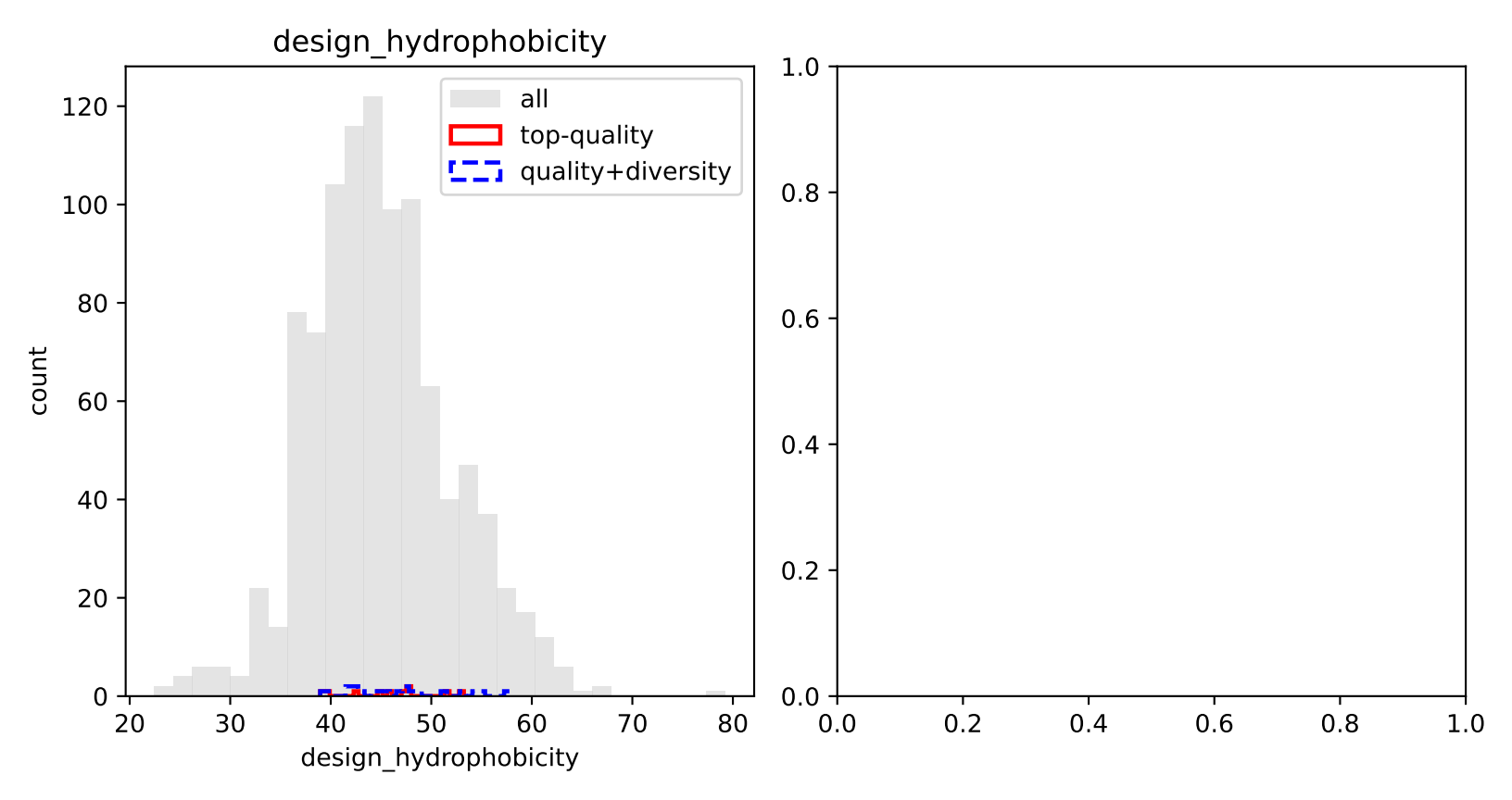

min_design_to_target_pae
all
top-quality
quality+diversity
count
min_design_to_target_pae
0
20
40
60
80
5
10
15
20
25
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0


plip_saltbridge_refolded
all
top-quality
quality+diversity
count
plip_saltbridge_refolded
0
25
50
75
100
125
150
175
0
2
4
6
8
10
12
14
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0


plip_hbonds_refolded
all
top-quality
quality+diversity
count
plip_hbonds_refolded
0
25
50
75
100
125
150
175
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0
0.2
0.4
0.6
0.8
1.0


delta_sasa_refolded
all
top-quality
quality+diversity
count
delta_sasa_refolded
0
20
40
60
80
500
1000
1500
2000
2500
0.0
0.2
0.4
0.6
0.8
1.0


design_largest_hydrophobic_patch_refolded
all
top-quality
quality+diversity
count
0
20
40
60
80
100
120
0
500
1000
1500
2000
2500
3000
3500
design_largest_hydrophobic_patch_refolded
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0


design_hydrophobicity
all
top-quality
quality+diversity
count
design_hydrophobicity
0
20
40
60
80
100
120
20
30
40
50
60
70
80
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0